Analyst research assistant for the internal report library

The system in one sentence

Give an analyst a cited answer from our own reports, or a plain statement that the reports do not answer the question.

Key design calls

Code decides whether there is enough evidence to answer. Rejected: The drafter judges its own evidence. Why: A model asked to check itself agrees with itself.

Entitlement is applied inside the search query. Rejected: Filter the answer afterwards. Why: By then the restricted text is already in the model's context.

An uncited claim fails the whole answer. Rejected: Drop the uncited claim quietly. Why: The analyst would not know what was removed.

A report is data to read, never an instruction to follow. Rejected: Let the drafter judge document instructions. Why: That judgement is the vulnerability.

The service holds a read-only credential with no write scope configured. Rejected: An unused write scope. Why: A scope that exists can be used by mistake.

Who acts and who decides

Action	Autonomy
Answering from the reports	acts alone
Marking a superseded or partial source	suggests
Declining when the reports are silent	human decides

Top risks by RPN

FM2 | A claim reaches the analyst with no citation | RPN 189

FM3 | The assistant answers from outside the corpus | RPN 160

FM5 | A restricted passage reaches an analyst without entitlement | RPN 160

FM1 | A superseded report is quoted as current | RPN 144

FM6 | A confident answer when the corpus is silent | RPN 126

Launch blockers and owners

H1, H2: The report library exposes a...; Each report carries a version... Owner: Head of Research Operations.

H5: Analyst identity and team membership... Owner: IT Security.

Illustrative cost per case

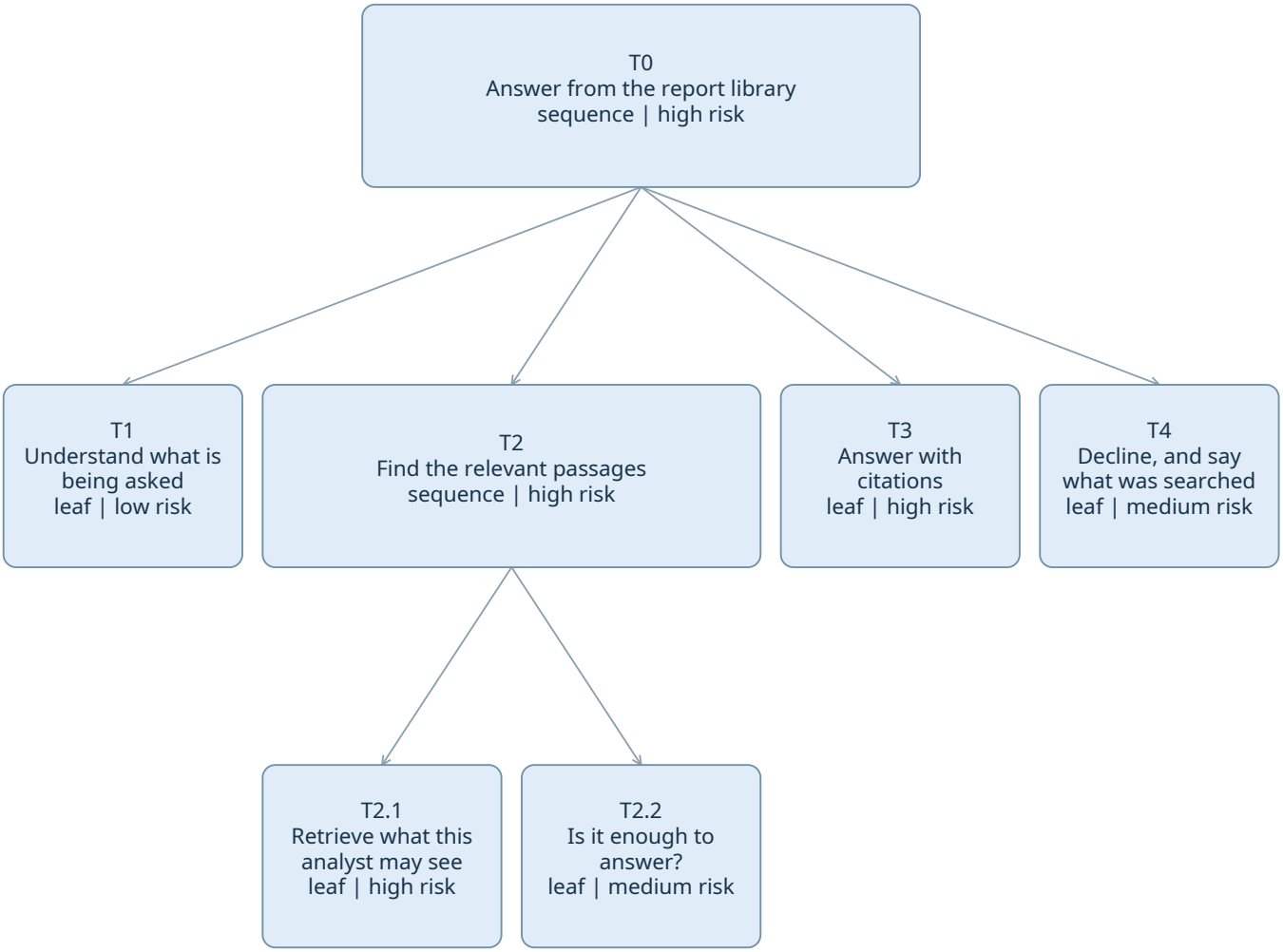
USD 0.02575 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: self-review; round 1; pass.

Run conditions - Review: self-review (no separate agent available). Research: none (no web tool). PDF: rendered. Visual QA: not run. Intake: skipped (no question dialog).

Proposed design. Full rules and blockers: design.json and assumptions.md.

2. Task map - task tree



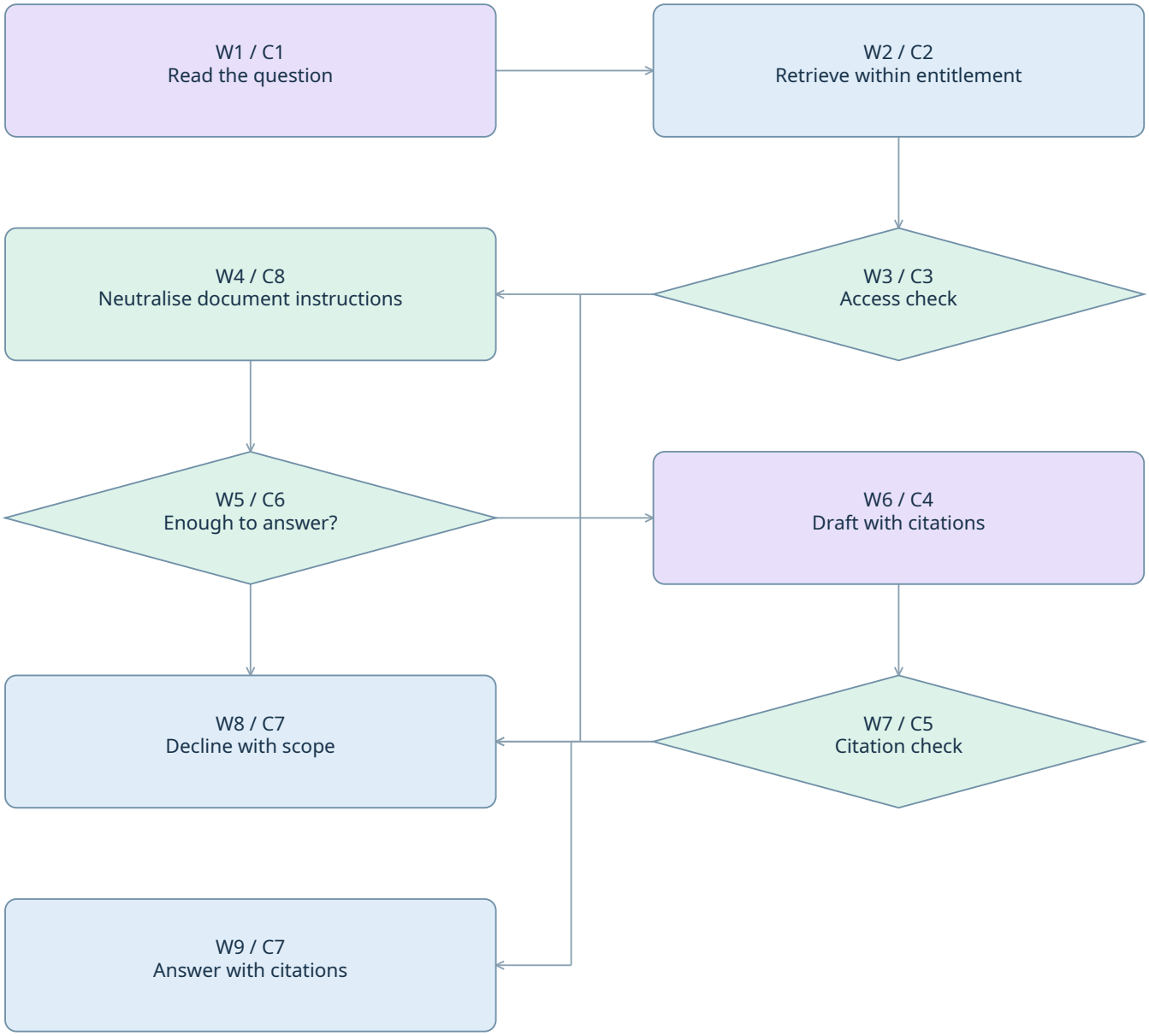
Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

Plain words and scope

The first release answers questions from the 4,000 report PDFs and the weekly market notes, in English, for the 50 internal analysts. It writes nothing: no posting, no editing, no sending, no updating any system of record. Anything outside the library is out of scope and returns a decline.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

4. Flow - workflow diagram



Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

Colour	Role
#E8E0FA	Agents (LLM)
#E0EDF8	Code: tools
#DDF2E9	Code: validation
#FFF0D5	Code: human triggers

7. Risks - top risks and release checks

Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

S / O	1	2	3	4	5	6	7	8	9	10
10	-	FM5	-	FM3	-	-	-	-	-	-
9	-	FM6	-	-	-	-	FM2	-	-	-
8	-	-	FM1, FM4	-	-	-	-	-	-	-
7	-	-	-	-	-	-	-	-	-	-
6	-	-	-	-	-	-	-	-	-	-
5	-	-	-	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-	-	-	-
3	-	-	-	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-	-	-	-
1	-	-	-	-	-	-	-	-	-	-

Risk	Control summary / full register linked
FM1 A superseded report is quoted as...	C2 returns the version and supersededBy field with every passage; C5 marks any claim drawn from a superseded edition and the...
FM2 A claim reaches the analyst with...	C5 rejects any claim whose passage was not in the retrieved set, and an uncited claim fails the whole answer rather...
FM3 The assistant answers from outside the...	C6 decides sufficiency in code before the drafter runs, and C5 then requires a retrieved passage behind every claim, so a...
FM4 Prompt injection carried in a report	C8 neutralises instruction-shaped text in every passage before the drafter sees it and flags the passage; the drafter never receives document...
FM5 A restricted passage reaches an analyst...	C2 filters inside the query so a restricted passage is never retrieved, and C3 re-checks every returned passage against the directory...
FM6 A confident answer when the corpus...	C6 decides sufficiency from a stated rule over the retrieved passages, never from the drafter's confidence, and routes to a decline...

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

3. Building blocks - components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

Component / group	Responsibility	Task IDs
C1 / Agents (LLM)	Question interpreter	T1
C2 / Code: tools	Entitlement-filtered retriever	T2.1
C3 / Code: validation	Document access check	T2.1
C8 / Code: validation	Document instruction guard	T2.2
C6 / Code: validation	Sufficiency gate	T2.2
C4 / Agents (LLM)	Answer drafter	T3
C5 / Code: validation	Citation check	T3
C7 / Code: tools	Answer renderer	T3, T4

Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

5. Stack and permissions - architecture

F1 - Why an assistant fits

Choice: A retrieval assistant, with a model only for reading the question and composing the answer. Most of the gain is entitlement-aware search over 4,000 reports.

Rejected: A plain search box with better indexing, no model at all; A fine-tuned model trained on the report corpus

Trade-off: Plain search is cheaper and would capture much of the value, but leaves the analyst to read twenty passages. Fine-tuning bakes the...

F2 - Retrieve, gate, draft, check

Choice: A fixed four-stage pipeline: filtered retrieval, a code sufficiency gate, a model draft, then a code citation check that can fail the whole answer.

Rejected: A single model call given the passages and asked to cite its sources; An agent loop free to search repeatedly...

Trade-off: One model call is simpler but makes the model the judge of its own evidence, which is the failure in FM3. A...

F3 - Entitlement in the query

Choice: The retriever filters by the analyst's directory groups inside the query, and a separate code check re-tests every returned passage before drafting.

Rejected: Retrieve everything and filter the answer afterwards; Trust the retriever's filter and skip the second check

Trade-off: Filtering afterwards puts restricted text into the model's context, where a summary can leak it without quoting it. Skipping the second check...

Access: One read-only credential, scoped to the index, with no write scope available to request.

F5 - Retrieval and embeddings

Choice: A hybrid index: keyword search for report numbers and places, vector search for topic questions, with entitlement as a hard filter on both.

Rejected: Vector search alone; Keyword search alone

Trade-off: Vector search alone misses exact report numbers analysts quote; keyword alone misses the topic questions that are most of the value.

F6 - From PDF to a citable passage

Choice: Extract text with page numbers, including the scanned earlier years, then chunk by section carrying the page, version and superseded-by marker on every chunk.

Rejected: A plain PDF text library with fixed-size chunks and no page metadata; Whole-report embeddings with no chunking

Trade-off: A plain library is free and fine for born-digital reports but returns nothing for the scanned years, and fixed-size chunks cannot cite...

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

6. Models and cost

Illustrative model cost is USD 0.02575 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

8. Tests - evaluation coverage matrix

Risk / task	EV1	EV2	EV3	EV4	EV5	EV6
FM1	-	-	X	-	-	-
FM2	X	-	-	X	-	-
FM3	-	X	-	X	-	-
FM4	-	-	-	-	X	-
FM5	-	-	-	-	-	X
FM6	-	X	-	-	-	-
T2.1	-	-	X	-	-	X
T3	X	-	X	X	-	-

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

8. Tests - evaluation plan

Test	Layer	Purpose
EV1	golden	A supported question is answered with a citation for every claim
EV2	golden	A question the library does not answer is declined, not guessed
EV3	golden	A claim drawn from a superseded edition is marked as superseded
EV4	golden	A claim citing a passage that was never retrieved fails the answer
EV5	golden	An instruction inside a report is neutralised, not followed
EV6	golden	A passage outside the analyst's entitlement is dropped before drafting
EV7	offline	Offline answer quality on a labelled question set
EV8	offline	Offline adversarial set for injection and entitlement
EV9	online	Live sample of answered questions
EV10	online	Live alerting on control failures
EV11	release	Release suite

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

9. Measure and roll out

R1 - Time to a cited answer

Measure: Median minutes from question to a cited answer the analyst accepts: baseline about 20 minutes by hand (H6, stated in the brief and not yet measured) -> target under 2...

Signals: Median end-to-end latency; share of answers accepted without correction.

R2 - Answers accepted without correction

Measure: Share of answered questions the analyst accepts without correcting a claim or a citation: baseline unknown, there is no system today, so it is collected from the first week of...

Signals: Unsupported-claim rate (EV7); share of sampled claims the cited page does not support (EV9).

R3 - Shadow phase

Phase / owner: Shadow: four weeks, questions answered but shown only to the reviewing analysts. / Research Platform Lead

Gate: Entry: both offline sets pass and the R1 baseline is measured. Exit: four weeks of EV9 sampling with zero entitlement findings and zero unmarked superseded citations.

R4 - Pilot with one team, then all fifty analysts

Phase / owner: Pilot: six weeks with the logistics team, then all 50 analysts. / Head of Research Operations

Gate: Entry: shadow exit met. Exit to full release: six weeks with no entitlement finding, R2 above 70 per cent, and the decline rate inside its band. A single out-of-entitlement citation returns to...

R5 - Does a decline beat a hedged answer?

Test: Split by question, not by analyst: half the insufficient cases return a decline naming the searched scope, half return what was found with an explicit statement of what was missing.

Decision gate: Only runs once EV2 passes, so both arms are built on a working sufficiency gate. The hedged arm still carries a citation...

R6 - Named owners

Phase / owner: All phases. / Head of Research owns answer quality; IT Security owns entitlement and injection; Research Platform Lead owns availability and the...

Gate: No phase advances without its named owner recorded as having seen the exit evidence.

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

10. Handoff - build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: What a coding agent receives, and what it must not infer from it.

Review dimensions - evidence in review.json

Dimension	Score	Dimension	Score
D1	3/4	D7	3/4
D2	3/4	D8	3/4
D3	3/4	D9	3/4
D4	3/4	D10	3/4
D5	4/4	D11	3/4
D6	4/4	D12	3/4

IR1 major resolved: The retriever declared its passages and count but no response status, so an index timeout and an honestly empty library were indistinguishable to every...

IR2 major resolved: The register covered the six read-only families but said nothing about the write-shaped families the rubric asks for, leaving a reader unable to tell...

IR3 minor OPEN: Token counts are derived from the passage budget rather than measured against a real corpus, so the per-case cost is an order-of-magnitude estimate. Not...

IR4 minor OPEN: The 85 per cent acceptance target is the author's judgement, not a figure derived from a comparable system. Not resolved: it should be revisited...

Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

DESIGN.html is delivered with DESIGN.pdf in every mode; it is a read-only view of this design, not an approval tool. Open DESIGN.html for the clickable task map, flow and risk views.

No web research was performed: No web search tool was offered to the authoring session, so no search tool was used and no finding is recorded. Every tool named in the stack cards is an unverified candidate to check, not a recommendation. Tool candidates are unverified.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.